Fiscal Year:	FY 2014	**Task Last Updated:**	FY 02/07/2014
PI Name:	German, Randall M. Ph.D.		
Project Title:	Multi-Scale Modeling and Experimentation on Liquid Phase Sintering in Gravity and Microgravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	randgerman@gmail.com	**Fax:**	FY none
PI Organization Type:	UNIVERSITY	**Phone:**	858-922-4985
Organization Name:	San Diego State University		
PI Address 1:	College of Engineering		
PI Address 2:	282 Surfview Court		
PI Web Page:			
City:	Del Mar	**State:**	CA
Zip Code:	92014	**Congressional District:**	53
Comments:	co-investigator Eugene A Olevsky, SDSU, Elisa Torresani, SDSU, MS 1326		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	10/01/2010	**End Date:**	11/30/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	Pending NOTE: End date changed to 11/30/2013 (previously was 3/30/2014) per NSSC information (Ed., 1/28/14) NOTE: End date changed to 3/31/2014 per NSSC information (Ed., 6/18/13)		
Key Personnel Changes/Previous PI:	November 2012. Jose Alvarado joined as visiting professor; Aditya Bothate (MS) went to work, Timothy Young took over the experimental work, and in December 2013 he graduated and took employment. Dr. Wei Li completed his PhD in December 2013. The undergraduate student Amber Ritchey also took full-time employment.		
COI Name (Institution):	Olevsky, Eugene (San Diego State University)		
Grant/Contract No.:	NNX10AV38G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A three year research project was completed on the gravity-dependent microstructure development and the related macroscopic shape evolution of multiphase powder materials during liquid phase sintering (LPS). The research achieved two critical goals: (i) in-depth analysis of the LPS induced pore-grain structure evolution by the de-convolution of the impact of gravity, and (ii) exploration of how sintering might be performed under microgravity conditions on the International Space Station as a means toward in-space fabrication and repair. Indeed, future NASA efforts to extend human exploration back to the Moon and beyond will require development of processes that permit fabrication under reduced gravity conditions. This capability is needed to reduce resource requirements for extraterrestrial construction and in space repair. Many proposed in-space fabrication and repair techniques involve particulate materials processing, such as via sintering, but limited data are available to assess the viability of these proposed net-shape techniques under reduced gravity. The research team developed a computer code within existing multiple physics simulation framework to attack the multi-scale simulation of LPS that to embrace both microstructure development and macroscopic shape distortion. Liquid phase sintering experiments on Earth and in microgravity will be conducted for a variety of tungsten heavy alloys to validate the sintering model framework. The research outcomes provide current and future space exploration information, such as needed for habitat creation, extraterrestrial exploration, and vehicle repair activities during trips to Mars or the Moon.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research is focused on liquid phase sintering, one of the most widely used manufacturing routes for ceramics, cemented carbides, powder metals, and composites. In the USA this is probably evaluated at $25 billion in annual product for metal cutting, oil well drilling, automotive engines, bearings, computers, and even watches. These components are designed for high performance and high precision, and are often fabricated in large production quantities. The net-shape fabrication, where final size and shape is achieved without machining, is a challenge since often dimensions change 15 to 20% due to shrinkage in going from the powder to the dense component. Thus, computer models of the sintering process are vital to proper tool and process design. Further, to attain specific mechanical, thermal, electrical, or other properties requires further attention to microstructure control. This research provides a computational model that predicts final size and shape and with multiple scale model predicts properties. Further, with robust inclusion of gravitational gradients, liquid-solid migration, and substrate friction, it will be useful for predicting component attributes in a range of gravitational conditions as required for eventual fabrication on the moon, planets, or microgravity; for example, in building structure or repairing vehicles.
Task Progress:	Much of the pending research came to completion with significant outputs. The experimental work successfully fabricated a range of dihedral angle and liquid-solid ratio tungsten heavy alloys for both ground and microgravity processing. Detailed laboratory experiments were conducted to isolate time, temperature, heating rate, and sample preparation conditions. The findings were melded into definition of experimental hardware that fits with the International Space Station furnace and that effort has progressed to full preliminary drawings and vendor identification. The matrix of experimental alloys, time, temperature, and other execution parameters are set and defended in terms of a successful science requirements review. In terms of the modeling efforts, the concepts were developed and embedded in existing multiple scale modeling frameworks, and data taken from various experiments were used to validate the approach. This has drawn the laboratory and experimental efforts into a convergence where samples and experiments and measures are defined for simulation. The next round of work will move to actual flight hardware (ground based) and expansion of the simulation to include pore migration in the viscous liquid-solid mixture. Assuming satisfactory results, the experiments will be ready for launch with an anticipated flight in June 2015. NOTE: project continues as "Multi-Scale Modeling and Experimentation on Liquid Phase Sintering in Gravity and Microgravity Environments (NNX14AB31G)."
Bibliography Type:	Description: (Last Updated: 03/27/2025)
Articles in Peer-reviewed Journals	Alvarado-Contreras JA, Olevsky EA, Maximenko AL, German RM. "A continuum approach for modeling gravitational effects on grain settling and shape distortion during liquid phase sintering of tungsten heavy alloys." Acta Materialia, 2014 Feb;65:176-84. http://dx.doi.org/10.1016/j.actamat.2013.10.059 , Feb-2014
Articles in Peer-reviewed Journals	Alvarado-Contreras JA, German RM, Maximenko AL, Olevsky EA. "Coupled densification-shape distortion analysis of liquid phase sintering affected by gravity." Metallurgical and Materials Transactions A. 2014 Feb;45(2):927-33. http://dx.doi.org/10.1007/s11661-013-2050-y , Feb-2014
Articles in Peer-reviewed Journals	Alvarado-Contreras JA, Olevsky EA, Maximenko AL, German RM. "Kinetics of shrinkage and shape evolution during liquid phase sintering of tungsten heavy alloy." Journal of Materials Science. 2014 Feb; 49(3):1130-7. http://dx.doi.org/10.1007/s10853-013-7791-x , Feb-2014
Articles in Peer-reviewed Journals	Alvarado-Contreras JA, Olevsky EA, German RM. "Modeling of gravity-induced shape distortions during sintering of cylindrical specimens." Mechanics Research Communications. 2013 Jun;50:8-11. http://dx.doi.org/10.1016/j.mechrescom.2013.02.007 , Jun-2013
Articles in Peer-reviewed Journals	Park SJ, Johnson JL, Wu Y, Kwon YS, Lee S, German RM. "Analysis of the effect of solubility on the densification behavior of tungsten heavy alloys using the master sintering curve approach." International Journal of Refractory Metals and Hard Materials. 2013 Mar;37:52-9. http://dx.doi.org/10.1016/j.ijrmhm.2012.10.016 , Mar-2013
Articles in Peer-reviewed Journals	German RM. "Coordination number changes during powder densification." Powder Technology. 2014 Feb;253:368-76. http://dx.doi.org/10.1016/j.powtec.2013.12.006 , Feb-2014
NASA Technical Documents	Olevsky EA, German RM. "Multi-Scale Modeling Of Liquid Phase Sintering Affected By Gravity: Preliminary Analysis." In: Materials Research in Microgravity 2012. R. Hyers, et al., Eds. Proceedings of a symposium held at the 141st TMS 2012 Annual Meeting and Exhibition, Orlando, Florida, March 11–15, 2012. Huntsville, AL : NASA Marshall Space Flight Center, 2012. NASA/CP—2012–217466. p. 113-119. http://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20120015582_2012018023.pdf , Sep-2012

Papers from Meeting Proceedings	Young TH, Wei L, Olevsky E, Whychell D, German RM. "Densification and Distortion of Tungsten Alloys using Copper-Nickel-Manganese." PM 2012 - 2012 Powder Metallurgy World Congress & Exhibition (PM2012), Yokohama, Japan, October 2012. Proceedings PM 2012 Yokohama Powder Metallurgy World Congress. Kyoto, Japan : Japan Society of Powder and Powder Metallurgy, 2013. Paper 16D-T3-7. , Jan-2013
Papers from Meeting Proceedings	Young TH, Wei L, Olevsky E, McKittrick J, German RM. "Densification and Distortion of Tungsten Alloys Using Low Sintering Temperatures." Powder Tech 2012. 2012 International Conference on Powder Metallurgy & Particulate Materials, Nashville, Tennessee, June 10–13, 2012. Advances in Powder Metallurgy & Particulate Materials—2012. Proceedings of the 2012 International Conference on Powder Metallurgy & Particulate Materials, June 10–13, Nashville, Tennessee. Princeton, NJ : Metal Powder Industries Federation, 2012, part 5, p. 34-44. , Jun-2012
Papers from Meeting Proceedings	Bothate A, German RM, Li W, Olevsky EA, Daoush WM, Moustafa S, Whychell D. "Advances in W-Cu: New Powder Systems." Eighth International Conference on Tungsten, Refractory & Hardmaterials, San Francisco, CA, May 18–21, 2011. In: Tungsten, Refractory & Hardmaterials VIII. Proceedings of the Eighth International Conference on Tungsten, Refractory & Hardmaterials. Princeton, NJ : Metal Powder Industries Federation, 2011. p. 7.6-7.20. ISBN: 978-0-9819496-7-3. , Jun-2011
Papers from Meeting Proceedings	German RM. "Self-Similar Microstructure and Property Trajectories for Sintering." 18th Plansee Seminar – International Conference on Refractory Metals and Hard Materials, Reutte, Austria, June 3-7, 2013. Proceedings 18th Plansee Seminar – International Conference on Refractory Metals and Hard Materials. Metallwerk Plansee, Reutte, Austria, 2013. Paper RM35, p. 345-356. , Jun-2013